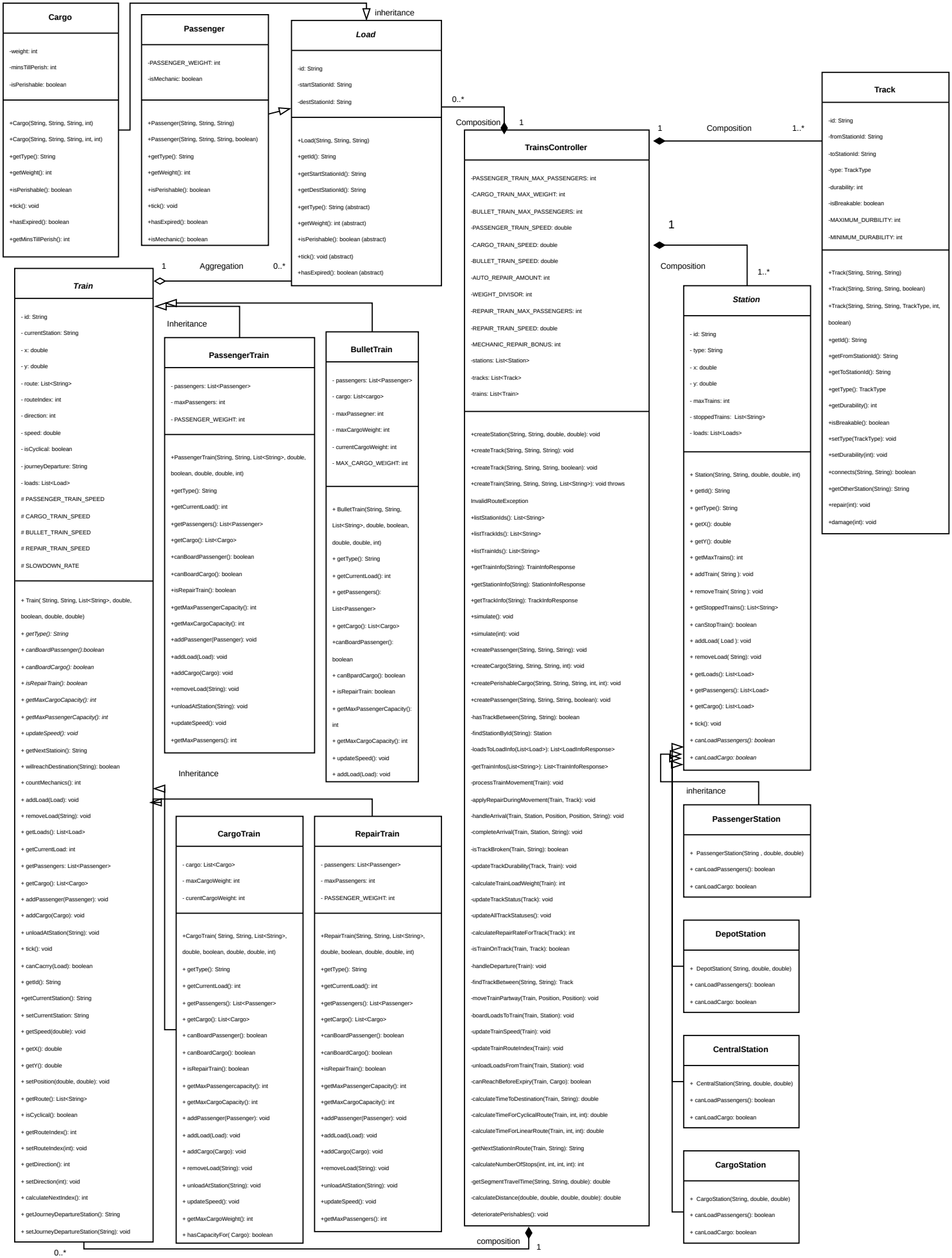

Cargo
-weight: int
-minsTillPerish: int
-isPerishable: boolean
+Cargo(String, String, String, int)
+Cargo(String, String, String, int, int)
+getType(): String
+getWeight(): int
+isPerishable(): boolean
+tick(): void
+hasExpired(): boolean
+getMinsTillPerish(): int
inheritance
Passenger
-PASSENGER_WEIGHT: int
-isMechanic: boolean
+Passenger(String, String, String)
+Passenger(String, String, String, boolean)
+getType(): String
+getWeight(): int
+isPerishable(): boolean
+tick(): void
+hasExpired(): boolean
+isMechanic(): boolean
Load
-id: String
-startStationId: String
-destStationId: String
+Load(String, String, String)
+getId(): String
+getStartStationId(): String
+getDestStationId(): String
+getType(): String (abstract)
+getWeight(): int (abstract)
+isPerishable(): boolean (abstract)
+tick(): void (abstract)
+hasExpired(): boolean (abstract)
0..*
Composition
1
TrainsController
-PASSENGER_TRAIN_MAX_PASSENGERS: int
-CARGO_TRAIN_MAX_WEIGHT: int
-BULLET_TRAIN_MAX_PASSENGERS: int
-PASSENGER_TRAIN_SPEED: double
-CARGO_TRAIN_SPEED: double
-BULLET_TRAIN_SPEED: double
-AUTO_REPAIR_AMOUNT: int
-WEIGHT_DIVISOR: int
-REPAIR_TRAIN_MAX_PASSENGERS: int
-REPAIR_TRAIN_SPEED: double
-MECHANIC_REPAIR_BONUS: int
-stations: List<Station>
-tracks: List<Track>
-trains: List<Train>
+createStation(String, String, double, double): void
+createTrack(String, String, String): void
+createTrack(String, String, String, boolean): void
+createTrain(String, String, String, List<String>): void throws InvalidRouteException
+listStationIds(): List<String>
+listTrackIds(): List<String>
+listTrainIds(): List<String>
+getTrainInfo(String): TrainInfoResponse
+getStationInfo(String): StationInfoResponse
+getTrackInfo(String): TrackInfoResponse
+simulate(): void
+simulate(int): void
+createPassenger(String, String, String): void
+createCargo(String, String, String, int): void
+createPerishableCargo(String, String, String, int, int): void
+createPassenger(String, String, String, boolean): void
-hasTrackBetween(String, String): boolean
-findStationById(String): Station
-loadsToLoadInfo(List<Load>): List<LoadInfoResponse>
-getTrainInfos(List<String>): List<TrainInfoResponse>
-processTrainMovement(Train): void
-applyRepairDuringMovement(Train, Track): void
-handleArrival(Train, Station, Position, Position, String): void
-completeArrival(Train, Station, String): void
-isTrackBroken(Train, String): boolean
-updateTrackDurability(Track, Train): void
-calculateTrainLoadWeight(Train): int
-updateTrackStatus(Track): void
-updateAllTrackStatuses(): void
-calculateRepairRateForTrack(Track): int
-isTrainOnTrack(Train, Track): boolean
-handleDeparture(Train): void
-findTrackBetween(String, String): Track
-moveTrainPartway(Train, Position, Position): void
-boardLoadsToTrain(Train, Station): void
-updateTrainSpeed(Train): void
-updateTrainRouteIndex(Train): void
-unloadLoadsFromTrain(Train, Station): void
-canReachBeforeExpiry(Train, Cargo): boolean
-calculateTimeToDestination(Train, String): double
-calculateTimeForCyclicalRoute(Train, int, int): double
-calculateTimeForLinearRoute(Train, int, int): double
-getNextStationInRoute(Train, String): String
-calculateNumberOfStops(int, int, int, int): int
-getSegmentTravelTime(String, String, double): double
-calculateDistance(double, double, double, double): double
-deteriorate Perishables(): void
1
Composition
1..*
Track
-id: String
-fromStationId: String
-toStationId: String
-type: TrackType
-durability: int
-isBreakable: boolean
-MAXIMUM_DURBILITY: int
-MINIMUM_DURABILITY: int
+Track(String, String, String)
+Track(String, String, String, boolean)
+Track(String, String, String, TrackType, int, boolean)
+getId(): String
+getFromStationId(): String
+getToStationId(): String
+getType(): TrackType
+getDurability(): int
+isBreakable(): boolean
+setType(TrackType): void
+setDurability(int): void
+connects(String, String): boolean
+getOtherStation(String): String
+repair(int): void
+damage(int): void
1
Composition
1..*
Station
- id: String
- type: String
- x: double
- y: double
- maxTrains: int
- stoppedTrains: List<String>
- loads: List<Loads>
+ Station(String, String, double, double, int)
+ getId(): String
+ getType(): String
+ getX(): double
+ getY(): double
+ getMaxTrains(): int
+ addTrain(String): void
+ removeTrain(String): void
+ getStoppedTrains(): List<String>
+ canStopTrain(): boolean
+ addLoad(Load): void
+ removeLoad(String): void
+ getLoads(): List<Load>
+ getPassengers(): List<Load>
+ getCargo(): List<Load>
+ tick(): void
+ canLoadPassengers(): boolean
+ canLoadCargo: boolean
inheritance
PassengerStation
+ PassengerStation(String , double, double)
+ canLoadPassengers(): boolean
+ canLoadCargo: boolean
DepotStation
+ DepotStation(String, double, double)
+ canLoadPassengers(): boolean
+ canLoadCargo: boolean
CentralStation
+ CentralStation(String, double, double)
+ canLoadPassengers(): boolean
+ canLoadCargo: boolean
CargoStation
+ CargoStation(String, double, double)
+ canLoadPassengers(): boolean
+ canLoadCargo: boolean
1
Aggregation
0..*
Train
- id: String
- currentStation: String
- x: double
- y: double
- route: List<String>
- routeIndex: int
- direction: int
- speed: double
- isCyclical: boolean
- journeyDeparture: String
- loads: List<Load>
PASSENGER_TRAIN_SPEED
CARGO_TRAIN_SPEED
BULLET_TRAIN_SPEED
REPAIR_TRAIN_SPEED
SLOWDOWN_RATE
+ Train(String, String, List<String>, double, boolean, double, double)
+ getType(): String
+ canBoardPassenger():boolean
+ canBoardCargo(): boolean
+ isRepairTrain(): boolean
+ getMaxCargoCapacity(): int
+ getMaxPassengerCapacity(): int
+ updateSpeed(): void
+ getNextStatioin(): String
+ willreachDestination(String): boolean
+ countMechanics(): int
+ addLoad(Load): void
+ removeLoad(String): void
+ getLoads(): List<Load>
+ getCurrentLoad: int
+ getPassengers: List<Passenger>
+ getCargo(): List<Cargo>
+ addPassenger(Passenger): void
+ addCargo(Cargo): void
+ unloadAtStation(String): void
+ tick(): void
+ canCacrry(Load): boolean
+ getId(): String
+getCurrentStation(): String
+ setCurrentStation: String
+ getSpeed(double): void
+ getX(): double
+ getY(): double
+ setPosition(double, double): void
+ getRoute(): List<String>
+ isCyclical(): boolean
+ getRouteIndex(): int
+ setRouteIndex(int): void
+ getDirection(): int
+ setDirection(int): void
+ calculateNextIndex(): int
+ getJourneyDepartureStation(): String
+ setJourneyDepartureStation(String): void
Inheritance
PassengerTrain
- passengers: List<Passenger>
- maxPassengers: int
- PASSENGER_WEIGHT: int
+PassengerTrain(String, String, List<String>, double, boolean, double, double, int)
+getType(): String
+getCurrentLoad(): int
+getPassengers(): List<Passenger>
+getCargo(): List<Cargo>
+canBoardPassenger(): boolean
+canBoardCargo(): boolean
+isRepairTrain(): boolean
+getMaxPassengerCapacity(): int
+getMaxCargoCapacity(): int
+addPassenger(Passenger): void
+addLoad(Load): void
+addCargo(Cargo): void
+removeLoad(String): void
+unloadAtStation(String): void
+updateSpeed(): void
+getMaxPassengers(): int
BulletTrain
- passengers: List<Passenger>
- cargo: List<cargo>
- maxPassegner: int
- maxCargoWeight: int
- currentCargoWeight: int
- MAX_CARGO_WEIGHT: int
+ BulletTrain(String, String, List<String>, double, boolean, double, double, int)
+ getType(): String
+ getCurrentLoad(): int
+ getPassengers(): List<Passenger>
+ getCargo(): List<Cargo>
+canBoardPassenger(): boolean
+ canBpardCargo(): boolean
+ isRepairTrain: boolean
+ getMaxPassengerCapacity(): int
+ getMaxCargoCapacity(): int
+ updateSpeed(): void
+ addLoad(Load): void
Inheritance
CargoTrain
- cargo: List<Cargo>
- maxCargoWeight: int
- curentCargoWeight: int
+CargoTrain(String, String, List<String>, double, boolean, double, double, int)
+ getType(): String
+ getCurrentLoad(): int
+ getPassengers(): List<Passenger>
+ getCargo(): List<Cargo>
+ canBoardPassenger(): boolean
+ canBoardCargo(): boolean
+ isRepairTrain(): boolean
+ getMaxPassengercapacity(): int
+ getMaxCargoCapacity(): int
+ addPassenger(Passenger): void
+ addLoad(Load): void
+ addCargo(Cargo): void
+ removeLoad(String): void
+ unloadAtStation(String): void
+ updateSpeed(): void
+ getMaxCargoWeight(): int
+ hasCapacityFor(Cargo): boolean
RepairTrain
- passengers: List<Passenger>
- maxPassengers: int
- PASSENGER_WEIGHT: int
+RepairTrain(String, String, List<String>, double, boolean, double, double, int)
+getType(): String
+getCurrentLoad(): int
+getPassengers(): List<Passenger>
+getCargo(): List<Cargo>
+canBoardPassenger(): boolean
+canBoardCargo(): boolean
+isRepairTrain(): boolean
+getMaxPassengerCapacity(): int
+getMaxCargoCapacity(): int
+addPassenger(Passenger): void
+addLoad(Load): void
+addCargo(Cargo): void
+removeLoad(String): void
+unloadAtStation(String): void
+updateSpeed(): void
+getMaxPassengers(): int
0..*
composition
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